Electrochimica Acta 107 (2013) 26–32

Contents lists available at ScienceDirect

Electrochimica Acta

journal homepage: www.elsevier.com/locate/electacta

Blends of lithium bis(oxalato)borate and lithium tetrafluoroborate: Useful substitutes for lithium difluoro(oxalato)borate in electrolytes for lithium metal based secondary batteries?

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ARTICLE INFO

Article history:
Received 23 March 2013
Received in revised form 3 May 2013
Accepted 28 May 2013
Available online 12 June 2013

Keywords:
Lithium metal anodes
Lithium difluoro(oxalato)borate
Lithium tetrafluoroborate
Lithium bis(oxalato)borate
SEI-formation and analysis
Coulombic efficiency

ABSTRACT

This work was inspired by the observation that some borates exchange their ligands that are attached to boron. Therefore, we investigated the effect of blends of two salts, lithium bis(oxalato)borate (LiBOB), lithium tetrafluoroborate ($LiBF_4$) on lithium cycling on copper with solutions based on ethylene carbonate (EC) and diethyl carbonate (DEC) (3:7, by wt.). Coulombic efficiencies of dissolution rate (*D*-rate) tests demonstrated an enhanced performance of LiBOB/$LiBF_4$ blend electrolytes compared to the LiBOB-based and $LiBF_4$-based electrolytes, increasing with increasing $LiBF_4$ content. The coulombic efficiencies of the LiBOB/$LiBF_4$ blend electrolyte with highest $LiBF_4$ content reached almost the coulombic efficiencies of the lithium difluoro(oxalato)borate (LiDFOB) based electrolyte at high current densities in *D*-rate tests. Voltage drop values, conductivity measurements, and AC impedance measurements show the good performance of the LiBOB/$LiBF_4$ blend electrolytes. The composition of the solid electrolyte interphase (SEI) of the LiBOB/$LiBF_4$ blend electrolytes studied by X-ray photoelectron spectroscopy (XPS) becomes more similar to the composition of the SEI of the LiDFOB-based electrolyte with increasing $LiBF_4$ content. Nuclear magnetic resonance (NMR) measurements showed that the formation of LiDFOB from $LiBF_4$ and LiBOB occurs already at room temperature by ligand exchange reactions at the boron atom. Therefore, we conclude that traces of LiDFOB in the LiBOB/$LiBF_4$ blend are responsible for the good performance of blended electrolytes.

1. Introduction

Lithium difluoro(oxalato)borate (LiDFOB) was recently investigated as a possible new salt for secondary lithium metal batteries [1]. Since this salt was first published by Zhang in 2006 [2] its characteristics have been extensively studied, mainly just for possible applications in secondary lithium ion batteries [3–7]. LiDFOB combines the advantages of the salts LiBOB and $LiBF_4$ when used in electrolytes for lithium ion cells as LiDFOB contains the same molecular moieties as lithium bis(oxalato)borate (LiBOB) and lithium tetrafluoroborate ($LiBF_4$) [2,3]. Therefore, LiDFOB-based electrolytes can fulfill the requirements for electrolytes needed for lithium ion batteries. The most important characteristic of LiDFOB is its ability to form a very stable and protecting solid electrolyte interphase (SEI [8,9]) on graphite electrodes, just like LiBOB [3,4]. This SEI in the presence of LiDFOB is a very good lithium ion conductor, comparable with the SEIs of $LiBF_4$-based electrolytes [3,4]. LiDFOB-based electrolytes have higher conductivity values than $LiBF_4$-based electrolytes [6]. Even more important than the conductivity of LiDFOB-based electrolytes is their higher cationic transference number. For 1 M LiDFOB in solvent blends of ethylene carbonate (EC) and diethyl carbonate (DEC) (3:7, by wt.), the cationic transference number is 0.33 and thus significantly higher than the cationic transference number for a comparable 1 M $LiPF_6$ in EC:DEC (3:7, by wt.) electrolyte, showing 0.24 only [10,11]. Furthermore, LiDFOB possesses better solubility in organic carbonates than LiBOB (1.4 mol $kg^{-1}{}_{solvent}$ vs. 0.65 mol $kg^{-1}{}_{solvent}$ in EC:DEC (3:7, by wt.)) [7,12]. The higher concentrated LiDFOB-based electrolytes show also a lower viscosity and better wettability than the comparable LiBOB-based electrolytes [3]. Other important features of LiDFOB include its very good protection of the aluminum current collector against anodic dissolution [6,7,13,14], its good thermal stability [5], and the absence of hydrogen fluoride (HF) generation upon hydrolysis [7,15,16]. This property could promote the cheaper and environmentally more preferable cathode materials such as lithium manganese oxide spinels [7] for lithium and lithium ion batteries [17,18].

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http://dx.doi.org/10.1016/j.electacta.2013.05.130

In addition, LiDFOB-based electrolytes show a better low temperature cycling performance than LiBOB-based electrolytes because of the reduced charge-transfer resistance [3,4] and a better lithium ion cell performance at high temperatures than with state of the art $LiPF_6$-based electrolytes [5,19]. The aim of this study was to investigate the influence of the BOB^--anion and the BF_4^--anion on the characteristics of the SEI on a lithium surface, especially when they are combined in one electrolyte. The results of the blend electrolytes were compared with the single salt electrolytes to separate the influences of the salts LiBOB and $LiBF_4$ in the blend on the behavior of the SEI and to learn more about the outstanding characteristics and behavior of the salt LiDFOB in plating/stripping experiments.

Cycling efficiencies of a dissolution rate (*D*-rate) test of lithium deposited on a copper substrate, which corresponds directly to the stripping rate of plated lithium, have been studied in electrolytes consisting of 1 M LiBOB:$LiBF_4$ (at molar ratios 1:1, 1:2, and 1:3) in the solvent blend EC:DEC (3:7, by wt.) that was also used in all other experiments for this study.

Results were compared with the results of the single salt electrolytes 0.5 M LiBOB, 1 M $LiBF_4$, and 1 M LiDFOB. Due to the low solubility of LiBOB in EC:DEC (3:7, by wt.) solvent mixtures it is not possible to get a 1 M LiBOB in EC:DEC (3:7, by wt.) electrolyte [7,12]. Of course the lower concentration of the LiBOB-based electrolyte will have an impact on the performance of this electrolyte, and possibly lower the coulombic efficiencies during the *D*-rate test compared to an imaginary 1 M LiBOB in EC:DEC (3:7, by wt.) electrolyte. To get more information about the influence of the SEI resistance and the electrolyte conductivity on the performance during the *D*-rate test, voltage drop values during the *D*-rate tests were compared with the conductivities of the electrolytes and AC impedance measurements of the lithium electrode-organic electrolyte interphase. For a better understanding of the SEI composition, the selected electrolytes were investigated by X-ray photoelectron spectroscopy (XPS). The blend electrolytes were also investigated by ^{11}B nuclear magnetic resonance (NMR) spectroscopy to study ligand exchange reactions entailing the formation of $DFOB^-$ anions [20], especially with regard to uncycled and cycled electrolytes, respectively. Finally, NMR studies extended over a period of 170 days were also performed at $DFOB^-$ containing salt and ionic liquids stored at 60 °C. The studied substances included the salt LiDFOB and $DFOB^-$ based ionic liquids with bulky cations such as tetraethylammonium (TEA^+), 1,3-ethylmethylimidazolium ($EMIM^+$), or 1,3-butylmethylimidazolium ($BMIM^+$). The solutions were analyzed 7 times during storage for several components, especially the change of the $DFOB^-$ content.

2. Experimental

2.1. Materials

The solvents EC (battery grade) and DEC (battery grade) were obtained from UBE Europe GmbH and used as received. LiBOB (battery grade) and $LiBF_4$ (battery grade) were purchased from Rockwood Lithium and Novolyte, respectively. LiDFOB was prepared with quantitative yield as already published elsewhere [21,22]. The electrolyte solutions consisting of 1 M LiBOB:$LiBF_4$ (at molar ratios 1:1, 1:2, and 1:3 in the solvent blend EC:DEC (3:7, by wt.)) were obtained by dissolving an adequate amount of the appropriate salts in the premixed solvent at 20 °C in a volumetric flask. The blend electrolytes 1 M LiBOB:$LiBF_4$ (1:n, by mole) in EC:DEC (3:7, by wt.) are denominated electrolyte A for $n = 1$, B for $n = 2$, and C for $n = 3$ throughout this publication. All solutions were prepared in an argon filled glove box (MBraun, LABmaster SP/DP glovebox workstation) with mass fractions of water and oxygen below 1 ppm. The water content in all electrolytes, analyzed by Karl Fischer Titration (Mitsubishi CA-200) with Aquamicron AKX (Mitsubishi) as analyte and Aquamicron CXU (Mitsubishi) as catholyte, was below 10 ppm. Lithium metal was purchased from Rockwood Lithium and copper foil was obtained from Evonik Industries. The copper foil was washed with water and ethanol and dried overnight in a glass oven (Büchi Labortechnik AG) at 60 °C in rotary vane pump vacuum at a pressure of about 0.1 Pa.

2.2. Methods

2.2.1. Cycling experiments

Lithium plating/stripping experiments were carried out in Swagelok®-cells (three electrode arrangement) with lithium as both reference (RE) and counter (CE) electrode [23]. The working electrode (WE) was a copper foil. The cells were assembled in the argon filled glove box. The lithium plating/stripping experiments were carried out at 20 ± 0.5 °C with a battery tester from MACCOR Inc., Series 4000. For *D*-rate (≙C-rate during discharge = Li dissolution = Li stripping) experiments the lithium plating current density was set to constant value of 0.1 mA cm^{-2}, whereas the lithium stripping current densities were varied, increasing from 0.1 mA cm^{-2} to 1 mA cm^{-2} in ten steps. The cut-off potentials were selected at 1.5 V vs. Li/Li^+ for lithium stripping and −0.3 V vs. Li/Li^+ for lithium plating. Lithium plating was conducted for one hour. The amount of lithium of the counter electrode was about 2000 times greater than the amount of lithium which was plated per cycle. This large excess of lithium was used in all laboratory cells.

2.2.2. Measurement of internal resistances of a cell

To determine the internal resistance of a cell we chose a fast method based on voltage measurements. The theoretical background and experimental setup have been previously described in detail [1,24–26].

2.2.3. Measurement of the specific conductivity of the electrolyte

The temperature dependence of the specific conductivity $\sigma/(S\,m^{-1})$ of the electrolytes was measured in the range from −40 °C to +60 °C with a Solartron 1260A impedance analyzer and a Solartron 1287A potentiostat from AMETEK GmbH. The experimental setup has been described in detail elsewhere [1].

2.2.4. Impedance measurements

The AC impedance at the lithium electrode-organic electrolyte interphase was measured with a Gamry Potentiostat Reference 600 and has been previously described in detail [1].

2.2.5. X-ray photoelectron spectroscopy (XPS)

For the XPS measurements, lithium discs were dipped in the different electrolytes and stored overnight in the glove box so that the electrolyte film could react with the lithium and build an appropriate SEI. The X-ray photoelectric investigations were carried out using an AXIS Ultra DLD (Kratos, UK) spectrometer. The samples were transferred from the argon glove box into the instrument without any air contact by using a sealed container. The samples were held there in vacuum overnight ($<5 \times 10^{-9}$ mbar). The X-ray gun was operated at 15 kV accelerating voltage and 12 meV emission current. A pass energy (PE) of 20 eV was used for all core scans. The spectra were calibrated due to the C 1s carbon at a binding energy of 284.8 eV.

2.2.6. Nuclear magnetic resonance (NMR)

^{11}B NMR spectra were recorded on an AVANCE III NMR instrument (Bruker) at 128.4 MHz for ^{11}B with 15% $BF_3{\cdot}Et_2O$ in $CDCl_3$ (0 ppm) as external standard. The NMR measurements were carried out for all blend electrolytes before and after the electrolyte

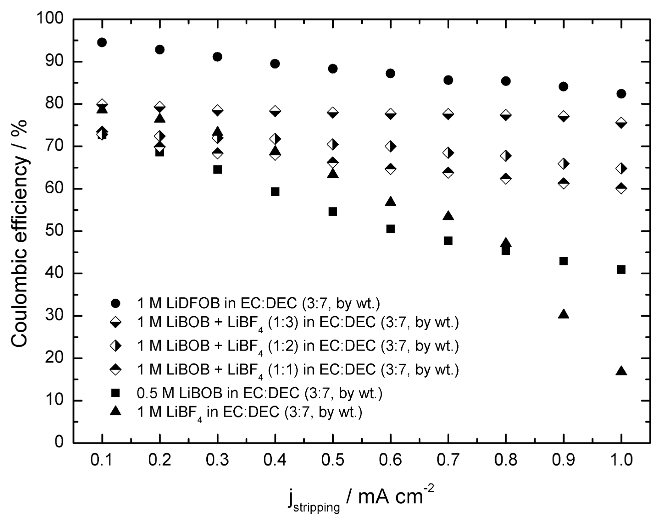

Fig. 1. Coulombic efficiencies of lithium plating/stripping experiments on copper substrate with increasing stripping current densities from 0.1 mA cm^{-2} to 1 mA cm^{-2} (*D*-rate test) for 1 M LiDFOB in EC:DEC (3:7, by wt.), 1 M LiBOB + $LiBF_4$ (1:3, by mole) in EC:DEC (3:7, by wt.), 1 M LiBOB + $LiBF_4$ (1:2, by mole) in EC:DEC (3:7, by wt.), 1 M LiBOB + $LiBF_4$ (1:1, by mole) in EC:DEC (3:7, by wt.), 0.5 M LiBOB in EC:DEC (3:7, by wt.), and 1 M $LiBF_4$ in EC:DEC (3:7, by wt.).

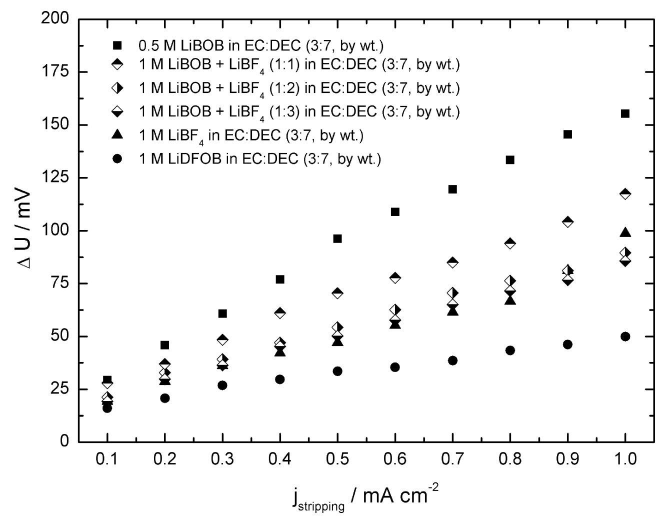

Fig. 2. Voltage drops during the *D*-rate tests for 0.5 M LiBOB in EC:DEC (3:7, by wt.), 1 M LiBOB + $LiBF_4$ (1:1, by mole) in EC:DEC (3:7, by wt.), 1 M LiBOB + $LiBF_4$ (1:2, by mole) in EC:DEC (3:7, by wt.), 1 M LiBOB + $LiBF_4$ (1:3, by mole) in EC:DEC (3:7, by wt.), 1 M $LiBF_4$ in EC:DEC (3:7, by wt.), and 1 M LiDFOB in EC:DEC (3:7, by wt.).

was cycled 20 times with a constant current i (i = 1 mA cm^{-2}). The solvent was dimethyl sulfoxide-d_6 (DMSO-d_6), 99.96 atom % *D* (Sigma–Aldrich).

3. Results and discussion

D-rate tests with constant plating current densities, $j_{plating}$, of 0.1 mA cm^{-2} for one hour and different stripping current densities, $j_{stripping}$, in the range from 0.1 mA cm^{-2} up to 1 mA cm^{-2} were performed. The results of the *D*-rate tests are shown in Fig. 1. All LiBOB/$LiBF_4$ blend electrolytes showed better results during the *D*-rate test than the single salt LiBOB- and $LiBF_4$-based electrolytes, respectively. The LiBOB-based electrolyte showed a linear decrease of coulombic efficiencies from 73 to 41% during the *D*-rate test. The $LiBF_4$-based electrolyte started with a higher coulombic efficiency of 79% with values decreasing similar to the blend electrolytes from j = 0.1 mA cm^{-2} to j = 0.4 mA cm^{-2}. In the range from j = 0.5 mA cm^{-2} to 0.8 mA cm^{-2} a stronger linear decrease was observed with the coulombic efficiencies of the $LiBF_4$-based electrolyte dropping below the efficiencies of the blend electrolytes. At the end of the *D*-rate test the coulombic efficiencies dropped significantly down to 17%. As a result, the $LiBF_4$-based electrolyte showed an even lower coulombic efficiency at the end of the *D*-rate test than the LiBOB-based electrolyte. The LiBOB/$LiBF_4$ blend electrolytes, however, had a much lower decrease of coulombic efficiencies during the *D*-rate test than the single salt electrolytes. The performance during the *D*-rate test was improved with increasing $LiBF_4$ content of the LiBOB/$LiBF_4$ salt mixtures. The electrolytes A and B started the *D*-rate test with about 73% of coulombic efficiency and finished the test after a linear decrease with 60 and 65% coulombic efficiency, respectively. Electrolyte C started the *D*-rate test with an even higher coulombic efficiency of 80% and ended with 76% after an only very slight decrease. The performances of the blend electrolytes during the *D*-rate test were also compared with the results of the *D*-rate test of the 1 M LiDFOB electrolyte (see also Fig. 1). The LiDFOB-based electrolyte showed the best performance during the *D*-rate test, with 95 to 82% coulombic efficiency. However, with increasing current densities the coulombic efficiencies of the LiDFOB-based electrolyte and the blend electrolyte C (with the highest $LiBF_4$-content) converged more and more. At a current density of 1 mA cm^{-2} the coulombic efficiencies of these two electrolytes differ by 6% only.

To sum up, the results of the *D*-rate tests showed clearly that mixtures of the salts LiBOB and $LiBF_4$ perform much better during the *D*-rate tests than the single salt electrolytes. The performance during the *D*-rate test of the blend electrolytes were improved with increasing $LiBF_4$ content and reached almost the coulombic efficiencies of the LiDFOB-based electrolyte, especially at high current densities. However, it is generally known that the coulombic efficiencies of lithium plating/stripping experiments cannot reach the coulombic efficiencies when graphite electrodes are cycled. According to Rauh et al. [27] the low coulombic efficiencies of lithium plating/stripping experiments can be explained by lithium dendrite growth and inevitable lithium encapsulation in the SEI. In both cases this lithium is lost for further cycling and causes the low coulombic efficiencies during the *D*-rate test.

To obtain more information about the influence of the cell resistances on the performance of the electrolytes in plating/stripping experiments, the voltage drops during the *D*-rate tests (see Fig. 2) were analyzed. As expected from the results of the *D*-rate test, the LiDFOB-based electrolyte showed the lowest ΔU values. The voltage drop values for the blend electrolytes increased with decreasing $LiBF_4$ content, which is also in compliance with the results of the *D*-rate test. Furthermore, the LiBOB-based electrolyte showed the highest ΔU values during the *D*-rate test. Besides, all electrolytes showed an increasing linear behavior of the voltage drop values with increasing current densities, as expected. The only electrolyte which did deviate from these results during the *D*-rate test (see Fig. 1) is the $LiBF_4$-based electrolyte. Despite the low coulombic efficiency values of the $LiBF_4$-based electrolyte, the ΔU values are even below the values of all blend electrolytes during the *D*-rate test from j = 0.1 mA cm^{-2} to j = 0.8 mA cm^{-2}. Although the voltage drop values of the $LiBF_4$-based electrolyte increased abruptly at the two highest current densities, these values are still below the voltage drop values of the electrolyte A and the LiBOB-based electrolyte.

The total resistance of typical laboratory cells is mainly determined by the resistance of the bulk electrolyte and the resistance of the SEI. To find an explanation for the performance of the investigated electrolytes during the *D*-rate test and the voltage drop values of this test, the two main resistances were investigated separately. Fig. 3 shows the results of the conductivity measurements. The $LiBF_4$-based electrolyte shows the smallest conductivity values.

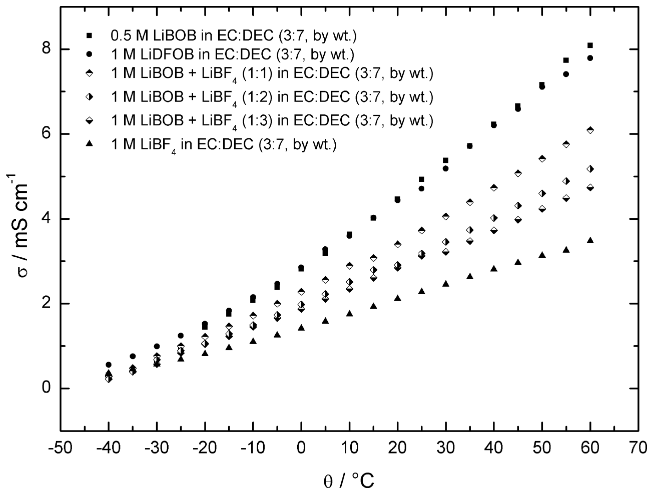

Fig. 3. Specific conductivities σ in the temperature range from −40 °C to 60 °C for 0.5 M LiBOB in EC:DEC (3:7, by wt.), 1 M LiDFOB in EC:DEC (3:7, by wt.), 1 M LiBOB + $LiBF_4$ (1:1, by mole) in EC:DEC (3:7, by wt.), 1 M LiBOB + $LiBF_4$ (1:2, by mole) in EC:DEC (3:7, by wt.), 1 M LiBOB + $LiBF_4$ (1:3, by mole) in EC:DEC (3:7, by wt.), and 1 M $LiBF_4$ in EC:DEC (3:7, by wt.) solutions.

This could be a reason for its poor performance during the *D*-rate test, but it cannot explain the very low voltage drop values. The conductivity of the blend electrolytes increased with decreasing $LiBF_4$ content while the voltage drop values of the blend electrolytes showed the opposite behavior at the *D*-rate test. The LiDFOB- and LiBOB-based electrolytes had almost the same conductivity values, whereas the LiDFOB-based electrolyte showed the lowest and the LiBOB-based electrolyte showed the highest voltage drop values during the *D*-rate test. Therefore, the resistances of the bulk electrolytes cannot be the explanation for the voltage drops during the *D*-rate tests of the investigated electrolytes.

To determine the resistance of the electrolytes' SEI, AC impedance measurements were performed, see Fig. 4. The *x*-axis intercept at high frequencies reflects the ohmic resistance of the bulk electrolyte [28], the slope of the impedance spectra at low frequencies is determined by diffusion processes in the electrolyte [28].

The Nyquist plots of the investigated electrolytes show clearly that the overall semicircle consists of several superimposed semi-

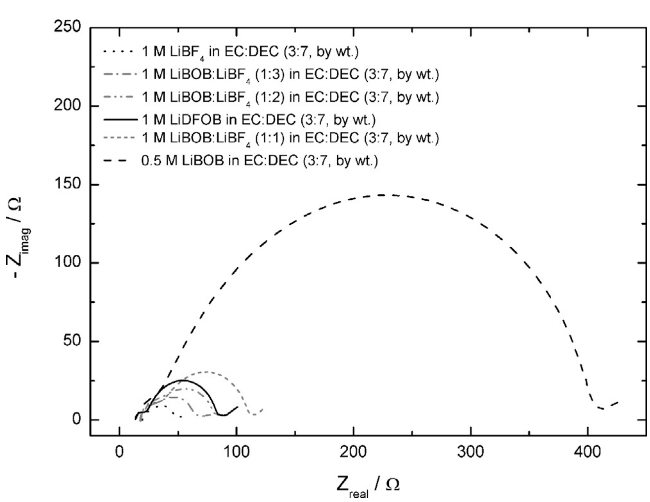

Fig. 4. Nyquist plot of 1 M $LiBF_4$ in EC:DEC (3:7, by wt.), 1 M LiBOB + $LiBF_4$ (1:3, by mole) in EC:DEC (3:7, by wt.), 1 M LiBOB + $LiBF_4$ (1:2, by mole) in EC:DEC (3:7, by wt.), 1 M LiDFOB in EC:DEC (3:7, by wt.) 1 M LiBOB + $LiBF_4$ (1:1, by mole) in EC:DEC (3:7, by wt.), and 0.5 M LiBOB in EC:DEC (3:7, by wt.).

circles. As generally accepted, the SEI consists of a multilayer structure [29] and therefore Aurbach et al. [30] suggested that the several superimposed semicircles represent the multilayer structure of the SEI on a lithium surface. The overall semicircle is related to several time constants [30], which can be assigned to the different single layers which built the multilayer structure of the SEI. Abe et al. [31,32] and Xu et al. [33,34] explained the existence of the mostly two pronounced superimposed semicircles with the so-called charge transfer resistance at lower frequencies and the process of Li^+ diffusion across SEI at higher frequencies. However, the separation and detailed discussion of the several time constants of the overall semicircle and its relation to various physical and chemical properties of the SEI, Li^+ diffusion and solvation sheath structure would be beyond the scope of this work. Therefore, for the following qualitative discussion just the diameter of the overall semicircle is taken into account because it corresponds to the main resistance of the SEI [30].

The diameters of the overall semicircle increased with decreasing $LiBF_4$ content, which is in good agreement with the coulombic efficiencies and voltage drop values of the *D*-rate tests. The LiBOB-based electrolyte showed the largest SEI resistance. Despite the high conductivity of the LiBOB-based electrolyte, the high resistance of the SEI determined its behavior during the *D*-rate test. The SEI resistance of the LiDFOB-based electrolyte is in the range of the resistances of the blend electrolytes. The superior behavior of the LiDFOB-based electrolyte during the *D*-rate test is therefore the result of its very conductive SEI combined with the high conductivity of the bulk electrolyte. The behavior of the $LiBF_4$-based electrolyte during the *D*-rate test also can be understood with the help of the impedance spectra. The $LiBF_4$-based electrolyte showed the smallest diameter of the overall semicircle which means that it had the least resistive SEI. This result can explain the low voltage drop values during the *D*-rate test. However, as known from literature, $LiBF_4$ forms not only a very conductive but also a rather unstable SEI [35]. Therefore, we suppose that the unstable SEI from $LiBF_4$-based electrolytes could be the reason for the low coulombic efficiencies, especially at high current densities. At low current densities the coulombic efficiencies are quite good and in the range of the blend electrolytes, despite the low conductivity of the $LiBF_4$-based electrolyte, because of the very conductive SEI of the $LiBF_4$-based electrolyte. However, we suppose that the SEI is more "stressed" at higher current densities and therefore suffers from more damages and plating/stripping processes of the $LiBF_4$-based electrolyte and its SEI components. This could lead to the serious drop of coulombic efficiencies at high current densities. In comparison, the LiBOB-based electrolyte forms a very stable but also rather thick and less conductive SEI, which also leads to a low performance during the *D*-rate test. Apparently, the blend electrolytes combine the advantages of the two salts, forming a very conductive and stable SEI which leads to the good results during the *D*-rate test. However, the lower conductivities of the blend electrolytes A to C compared to the LiDFOB-based electrolyte prevent the same excellent performance of the blend electrolytes during the *D*-rate test.

For a closer examination of the SEI composition of the LiBOB/$LiBF_4$ blend electrolytes on a lithium metal surface, Fig. 5 shows the B 1s spectra of all investigated electrolytes. The sampling depth of XPS is a few nm due to the mean free path of the electrons. A comparative XPS study of the salts LiBOB, $LiBF_4$ and LiDFOB in solutions with ethylene carbonate and ethylmethyl carbonate (EMC) (3:7, by wt.) to investigate the composition of the SEI on graphite electrodes was already published by Abraham et al. [36]. The results of this study are in good agreement with the results we obtained for the SEI compositions of electrolytes on a lithium metal surface, as shown in Fig. 5., including 0.5 M LiBOB in EC:DEC, 1 M $LiBF_4$, and 1 M LiDFOB, all in EC:DEC (3:7, by wt). The B 1s

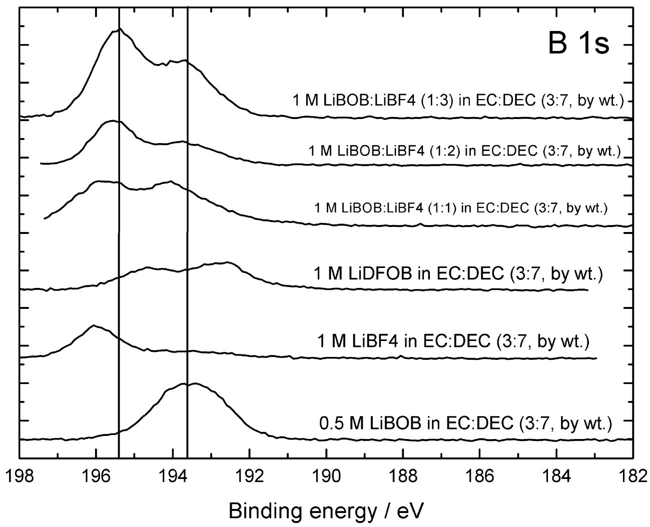

Fig. 5. B 1s XPS spectra of lithium metal discs after dipping in 0.5 M LiBOB in EC:DEC (3:7, by wt.), 1 M $LiBF_4$ in EC:DEC (3:7, by wt.), 1 M LiDFOB in EC:DEC (3:7, by wt.), 1 M LiBOB + $LiBF_4$ (1:1, by mole) in EC:DEC (3:7, by wt.), 1 M LiBOB + $LiBF_4$ (1:2, by mole) in EC:DEC (3:7, by wt.), and 1 M LiBOB + $LiBF_4$ (1:3, by mole) in EC:DEC (3:7, by wt.), respectively.

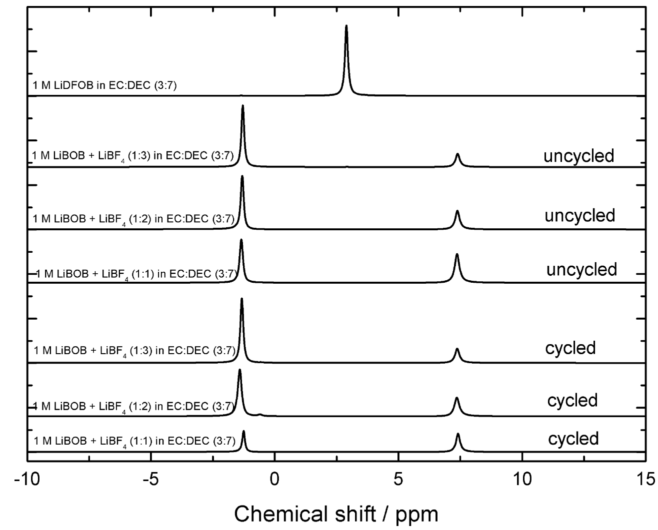

Fig. 6. ^{11}B NMR spectra of the three cycled (bottom) LiBOB/$LiBF_4$ blend electrolytes, 1 M LiBOB + $LiBF_4$ (1:1, by mole) in EC:DEC (3:7, by wt.), 1 M LiBOB + $LiBF_4$ (1:2, by mole) in EC:DEC (3:7, by wt.), and 1 M LiBOB + $LiBF_4$ (1:3, by mole) in EC:DEC (3:7, by wt.), the three uncycled (top) LiBOB/$LiBF_4$ blend electrolytes, 1 M LiBOB + $LiBF_4$ (1:1, by mole) in EC:DEC (3:7, by wt.), 1 M LiBOB + $LiBF_4$ (1:2, by mole) in EC:DEC (3:7, by wt.) and 1 M LiBOB + $LiBF_4$ (1:3, by mole) in EC:DEC (3:7, by wt.) and the 1 M LiDFOB in EC:DEC (3:7, by wt.) electrolyte.

spectrum of the LiBOB-based electrolyte shows one broad peak in the region of 192–195 eV. This peak is superimposed by the binding energies of the B 1s electrons of the unreacted LiBOB salt (located at about 194 eV) [36,37] and the binding energies of tri-coordinated borate species, such as B_2O_3 or H_3BO_3 [37] with the B 1s signal in the vicinity of 192–193 eV [36,37]. Also oligomers with tri-coordinated boron centers, which are the reduction products of the BOB^--anion with solvent molecules and which are supposed to play a major role in the composition of the SEI of LiBOB-based electrolytes, contribute to this peak in the B 1s spectrum [37]. The XPS spectrum of the $LiBF_4$-based electrolyte shows one B 1s signal with the center at 196 eV, which can be attributed to the unreacted $LiBF_4$ salt [36]. The reduction products of $LiBF_4$, such as Li_xBO_y and $Li_xBF_yO_z$, show a very slight signal in the region of 192–194 eV [36]. So we suppose that the reduction process of the salt $LiBF_4$ on a lithium surface is slower than the reduction processes of the salts LiBOB and LiDFOB. The small amount of $LiBF_4$-based reduction products may play a crucial role for the high conductivity of the SEI of $LiBF_4$-based electrolytes, but could also be the reason for the rather unstable SEI for this kind of electrolyte. The LiDFOB-based electrolyte shows two signals in the B 1s spectrum. The signal with the center at 194.5 eV can be assigned to the binding energy of the B 1s electrons of the unreacted LiDFOB salt [36]. The tri-coordinated borate species are shifted to smaller binding energies (192–194.5 eV) like the reduction products of the salt LiBOB [36]. The reduction products of the salt LiDFOB are Li_xBO_y or $Li_xBO_yF_z$ species [36] and we suppose that the B 1s electron binding energies of tri-coordinated boron centers of such fluorine-containing oligomeric species contribute to this signal. This oligomeric species can occur when the $DFOB^-$ anions react with solvent molecules, as proposed by Zhang [2,3]. The blend electrolytes also show two pronounced signals in the B 1s spectra. The signals at higher binding energies can be attributed to the B 1s electrons of the tetra-coordinated boron centers of the two salts LiBOB and $LiBF_4$, which form one superimposed peak. These peaks are centered between the $LiBF_4$ and LiBOB signals of the 1 M $LiBF_4$ and 0.5 M LiBOB electrolytes, respectively. But these peaks are still at higher binding energies than the signal of the LiDFOB salt. The second signal of the blend electrolytes at lower binding energies (192–194 eV) can be attributed to species with tri-coordinated boron centers. Possible compounds are H_3BO_3, Li_xBO_y and $Li_xBO_yF_z$ species, but also oligomeric species with tri-coordinated boron centers are possible. Interestingly, the centers of both signals of the LiBOB/$LiBF_4$ blend electrolytes shift to lower binding energies with increasing $LiBF_4$ content of the LiBOB/$LiBF_4$ blend electrolytes. Given that the two signals of the LiDFOB-based electrolyte are also shifted to lower binding energies compared to the signals of the single LiBOB- and $LiBF_4$-based electrolytes, the composition of the SEI of the LiBOB/$LiBF_4$ blend electrolytes became more similar to the composition of the SEI of the LiDFOB-based electrolyte with increasing $LiBF_4$ content. This shift of the SEI formed by LiBOB/$LiBF_4$ blend electrolytes to lower binding energies could be also influenced by the ligand exchange reaction between the salts LiBOB and $LiBF_4$ as shown by Lucht et al. at high temperatures [20]. Therefore, we suppose that by this reaction a small amount of LiDFOB could be formed and therefore also fluorine-containing oligomeric species in the SEI. Such a small amount of LiDFOB and fluorine-containing oligomeric SEI species in the blend electrolytes could also be a possibility for this slight shift of the signals to lower binding energies with higher $LiBF_4$-content. The B 1s spectra of the investigated electrolytes showed that the SEI composition of the blend electrolytes became more similar to the SEI composition of the LiDFOB-based electrolytes when the $LiBF_4$ content of the blend electrolytes was increased because of a shift to lower binding energies.

For a closer look to the behavior of the SEI formed by LiBOB/$LiBF_4$ blend electrolytes in the XPS measurements, NMR measurements were carried out to address the question whether the LiBOB/$LiBF_4$ blend electrolytes contained a small amount of the salt LiDFOB. To figure out whether LiDFOB might occur just during room temperature storage of the LiBOB/$LiBF_4$ blend electrolytes or due to the cycling of the cell, NMR measurements were carried out on LiBOB/$LiBF_4$ blend electrolytes before and after they were cycled 20 times in a cell. In Fig. 6 an overview of the ^{11}B NMR spectra of all LiBOB/$LiBF_4$ blend electrolytes before and after cycling and the spectrum of the LiDFOB-based electrolyte are presented for comparison. The singlet at 7.4 ppm can be assigned to the ^{11}B signal of the salt LiBOB and the peak at −1.3 ppm can be related to the ^{11}B signal of the salt $LiBF_4$ [7]. The ratio of the integrals is in accordance with the molar salt ratios of the blend electrolytes. The cycling of the electrolytes had no impact on the ratio of the integrals. Because of the stacked overview of the selected spectra there

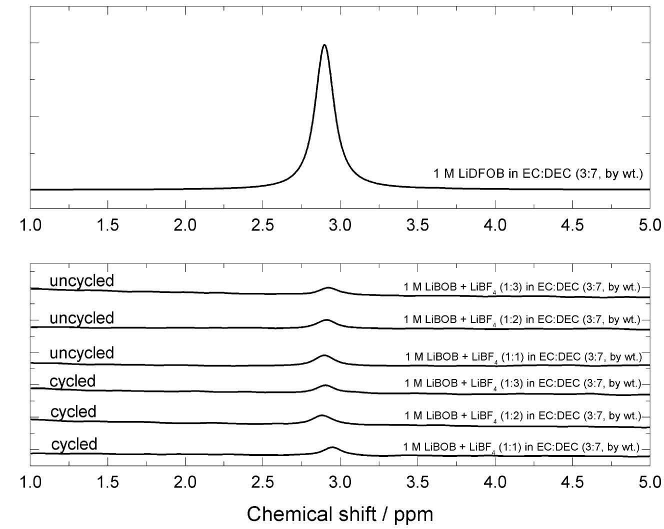

Fig. 7. Zoomed ^{11}B NMR spectra between 1–5 ppm. The spectrum at the top shows the 1 M LiDFOB in EC:DEC (3:7, by wt.) electrolyte. The spectra at the bottom show the three cycled LiBOB/$LiBF_4$ blend electrolytes, 1 M LiBOB + $LiBF_4$ (1:1, by mole) in EC:DEC (3:7, by wt.), 1 M LiBOB + $LiBF_4$ (1:2, by mole) in EC:DEC (3:7, by wt.), and 1 M LiBOB + $LiBF_4$ (1:3, by mole) in EC:DEC (3:7, by wt.) and the three uncycled LiBOB/$LiBF_4$ blend electrolytes, 1 M LiBOB + $LiBF_4$ (1:1, by mole) in EC:DEC (3:7, by wt.), 1 M LiBOB + $LiBF_4$ (1:2, by mole) in EC:DEC (3:7, by wt.), and 1 M LiBOB + $LiBF_4$ (1:3, by mole) in EC:DEC (3:7, by wt.).

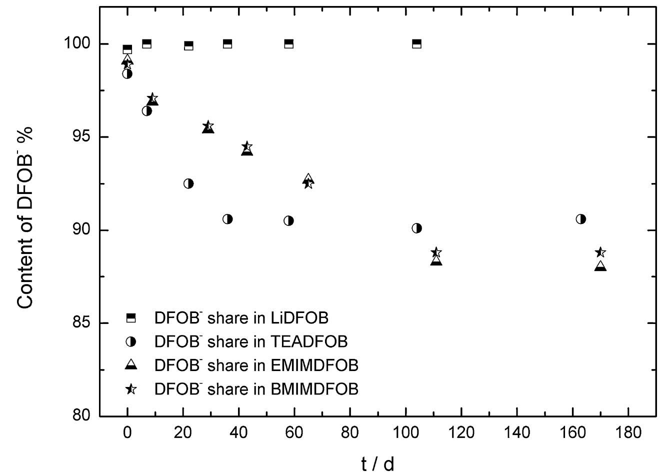

Fig. 8. Change of the DFOB- content in the salt LiDFOB and the ionic liquids TEADFOB, EMIMDFOB and BMIMDFOB, when starting from >98% DFOB- content and storing for prolonged time at 60 °C in argon.

is no ^{11}B signal from LiDFOB visible. Therefore, in Fig. 7 a zoomed view of the ^{11}B NMR spectra of the LiBOB/$LiBF_4$ blend electrolytes is shown at the bottom. At the top of the figure the ^{11}B NMR spectrum of the LiDFOB-based electrolyte is presented. As can be seen easily, all NMR spectra of the LiBOB/$LiBF_4$ blend electrolytes have a small and the ^{11}B NMR spectrum of the LiDFOB-based electrolyte shows a pronounced signal at 2.9 ppm which can be attributed to the anion $DFOB^-$ [7]. Thus, LiDFOB is obviously present in the investigated LiBOB/$LiBF_4$ blend electrolytes after some weeks of storage at room temperature. The occurrence of LiDFOB in the LiBOB/$LiBF_4$ blend electrolytes is not influenced by cycling of the electrolytes in a laboratory cell. Lucht et al. [20] also investigated the LiBOB, $LiBF_4$ and LiDFOB equilibrium via NMR measurements and discussed no LiDFOB signal in the ^{11}B NMR spectrum for their investigated LiBOB/$LiBF_4$ blend electrolyte, even after 6 months of storage at room temperature. However, in their published figure of the ^{11}B NMR experiment of a 0.5 M $LiBF_4$ + 0.5 M LiBOB in EC/EMC (3/7, v/v) after 6 months storage at room temperature, a small peak at about 2.5 ppm is visible, which could tentatively be assigned to traces of the salt LiDFOB [20]. The NMR measurements show that the LiBOB, $LiBF_4$ and LiDFOB equilibrium also exists at room temperature with the formation of traces of LiDFOB in LiBOB/$LiBF_4$ blend electrolytes. This is in accordance with the observation that the salt LiDFOB which initially had ca. 1–2% $LiBF_4$ and LiBOB as impurities after synthesis becomes increasingly pure over time if stored at 60 °C in argon atmosphere for few weeks. Finally, only LiDFOB can be detected by ^{11}B NMR which shows the tendency to form LiDFOB from $LiBF_4$ and LiBOB also in substance, especially with Li^+ as counter-cation. With soft, bulky organic cations such as TEA^+, $EMIM^+$, or $BMIM^+$ on the other hand, [21] this equilibrium is reached at about 88–90% $DFOB^-$, 5–6% $BF_4{}^-$, and 5–6% BOB^- for the corresponding $DFOB^-$-based salts/ionic liquids when starting from >98% $DFOB^-$ content and storing for prolonged time at 60 °C in argon. The change of the $DFOB^-$ content for prolonged storage time at 60 °C for the salt LiDFOB and the ionic liquids TEADFOB, EMIMDFOB and BMIMDFOB is shown in Fig. 8. It can be seen that the $DFOB^-$ content measured with ^{11}B NMR decreases in the ionic liquids, whereas it increases in LiDFOB. Thus, it can be rationalized that ligand exchange reactions lead to small traces of LiDFOB in 1 M LiBOB/$LiBF_4$ electrolyte solutions. Such traces of LiDFOB could play a role for the shift of the two signals of the LiBOB/$LiBF_4$ blend electrolytes to lower binding energies with enhanced $LiBF_4$ content in the B 1s spectra (see Fig. 5). The small amount of LiDFOB in the LiBOB/$LiBF_4$ blend electrolytes could therefore also play a crucial role for the good properties of the SEI of the LiBOB/$LiBF_4$ blend electrolytes and for their performance during the *D*-rate test.

4. Conclusions

To sum up, the results of this work show that salt mixtures of lithium bis(oxalato)borate (LiBOB) and lithium tetrafluoroborate ($LiBF_4$) in solutions based on ethylene carbonate (EC) and diethyl carbonate (DEC) (3:7, by wt.) show improved performance during lithium plating/stripping experiments on copper, especially with enhanced $LiBF_4$ content when compared to the single salt LiBOB- and $LiBF_4$-based electrolytes, respectively. Despite the good performance of the 1 M LiBOB:$LiBF_4$ (1:3, by mole) in EC:DEC (3:7, by wt.) electrolyte during the dissolution rate (*D*-rate) test, the performance of the 1 M lithium difluoro(oxalato)borate (LiDFOB) in EC:DEC (3:7, by wt.) electrolyte could not be fully reached. For the influence of the single salts LiBOB and $LiBF_4$ on the performance of the LiBOB/$LiBF_4$ blend electrolytes an extensive study of coulombic efficiencies and voltage drops during the *D*-rate tests and conductivity and AC impedance measurements were performed for all investigated electrolytes. The ligand exchange reaction of LiBOB with $LiBF_4$, yielding traces of LiDFOB is responsible for the good characteristics of the LiBOB/$LiBF_4$ blend electrolytes. The composition of the SEIs of the LiBOB/$LiBF_4$ blend electrolytes were investigated via XPS measurements and it was shown that with enhanced $LiBF_4$ content the SEI compositions of the LiBOB/$LiBF_4$ blend electrolytes became more similar to the SEI composition of the LiDFOB-based electrolyte. NMR measurements at solutions of said lithium salts show that the ligand exchange reactions also occur at room temperature with the formation of traces of LiDFOB in LiBOB/$LiBF_4$ blend electrolytes.

Acknowledgements

The authors gratefully acknowledge the financial support of this work by the DFG (German Research Foundation) as part of the research initiative PAK 177 “Funktionsmaterialien und Materialanalytik zu Lithium-Hochleistungsbatterien” (contract numbers

WI 2929/1-1 GO 1001-7-2, and WI 2929/5-1) on materials for lithium ion batteries and of the project “MEET-HiEnd”, project No. 03X4634A in the framework of the Excellent Battery Program of the German Ministry of Education and Science (BMBF).

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